PharmaLeaf · B.Pharm Semester I · PCI 2026 (NEP 2020)

BP101T Basics of Python Programming for Pharmaceutical Sciences

PharmaLeaf notes for B.Pharm Semester I (PCI 2026, NEP 2020). Written in easy language, unit by unit in syllabus order, with tables, diagrams and small programs. Every program in these notes was run, and the output shown is the real output.

Syllabus at a glance

BP101T is a 2-credit core theory course: 2 lectures per week, 30 hours in total, 50 marks (20 sessional + 30 end-semester).

Unit	Title	Topics	Hours
I	Introduction to Python programming	Installing Python and IDEs; variables, data types, type casting, operators; input/output; strings; standard and third-party libraries	6
II	Control structures and functions	if, if-else, if-elif-else, nested; for and while loops; break and continue; functions; dosage and BMI programs	6
III	Data structures and file handling	Lists, tuples, dictionaries; indexing and slicing; NumPy arrays; CSV files; healthcare datasets	6
IV	Data handling with Pandas	Series and DataFrame; reading CSV/Excel (PK and ADR data); head, tail, info, describe; cleaning; filtering; grouping	6
V	Data visualization with Matplotlib	Line, histogram, scatter, box plots; labels and legends; concentration-time curves, ADR rates, dissolution profiles; interpretation	6

How to use these notes

- Every program can be typed and run in Jupyter Notebook, PyCharm or VS Code.
- Lines starting with `#` inside a program are comments; Python ignores them.
- The box labelled Output shows exactly what the program prints.
- The two practice files `pk_study.csv` and `adr_reports.csv` are given in full in the Appendix at the end.

Unit I: Introduction to Python Programming (6 hours)

1.1 Introduction to Python

What is Python?

1. Python is a computer programming language. We use it to give step-by-step instructions to a computer.
2. It was created by Guido van Rossum and first released in 1991.
3. Its code looks almost like simple English, so it is easy for beginners.

Program and programming

1. A program is a set of instructions that tells the computer what to do, in the right order.
2. Writing these instructions is called programming (or coding).
3. Example: "Take the patient's weight, multiply it by 15, and show the dose" is a program written in plain English. Python lets us write the same thing in a form the computer understands.

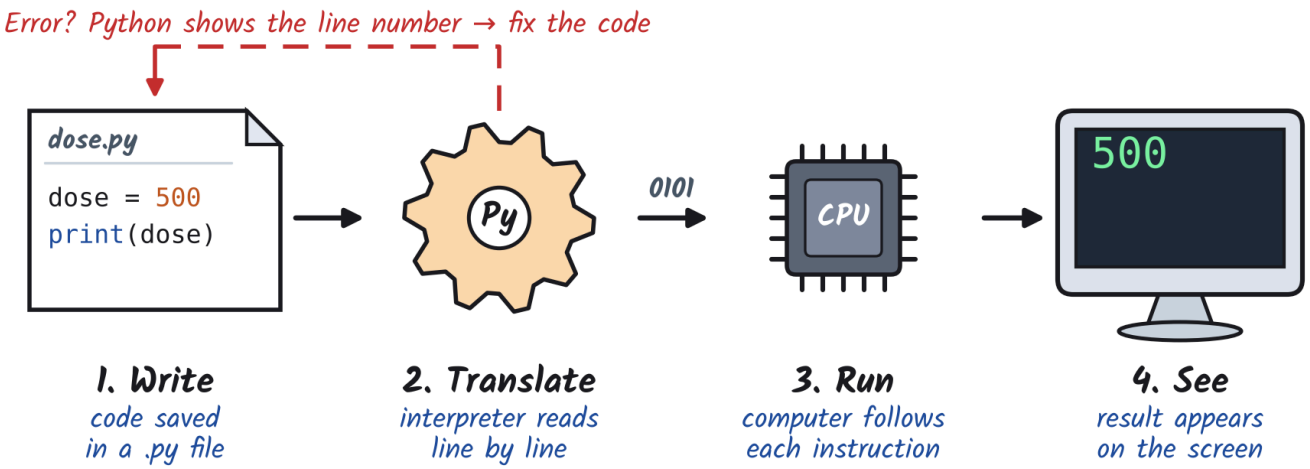
Features of Python

1. Easy to read and write: simple English-like words such as `print`, `if`, `and`.
2. Interpreted: Python runs the program line by line, so errors are easy to find.
3. Free and open source: anyone can download and use it.
4. Platform independent: the same program runs on Windows, Mac and Linux.
5. Dynamically typed: you do not need to declare the data type; Python finds it by itself.
6. Large library support: ready-made tools like NumPy, Pandas and Matplotlib save time.

How a Python program runs

1. You write the code and save it as a `.py` file (or type it in a notebook cell).
2. The Python interpreter reads the code one line at a time.
3. It translates each line into machine language that the computer understands.
4. The computer carries out the instruction and the result (output) appears on the screen.

How a Python Program Runs



Remember: Write → Translate → Run → See
Python is an INTERPRETED language: it translates and runs one line at a time.

PharmaLeaf

Uses of Python in pharmaceutical sciences

Area	How Python helps
Dose calculation	Calculates doses by body weight or body surface area quickly and without arithmetic mistakes
Data analysis	Handles large datasets of patients, drugs and sales using Pandas
Clinical trials	Organizes and compares trial results, finds averages and trends
Graphs and reports	Draws charts of drug stability, dissolution or sales using Matplotlib
Drug discovery	Studies chemical structures and predicts drug properties using special libraries
Automation	Repeats boring tasks (stock lists, label printing, reports) automatically

Your first Python program

```
print("Welcome to Pharmacy")
# Output: Welcome to Pharmacy
```

1.2 Installing Python and an IDE

A. Installing Python (Windows)

1. Open the official website python.org and go to Downloads.
2. Download the latest Python 3 installer for Windows.

3. Run the installer and tick the box "Add python.exe to PATH". This lets you run Python from any folder.
4. Click "Install Now" and wait until setup is complete.
5. Check the installation: open Command Prompt and type `python --version`. If it shows a version number (e.g., Python 3.12), Python is installed.

Easy option: install Anaconda. It gives Python, Jupyter Notebook, NumPy and Pandas together in one installation.

B. What is an IDE?

1. IDE stands for Integrated Development Environment.
2. It is one software that has everything needed to write, run and correct programs: a code editor, a run button, an output window and error-checking tools.
3. "Integrated" means all these tools are joined together in one place.

C. Popular Python IDEs

Feature	Jupyter Notebook	PyCharm	VS Code
Made by	Project Jupyter (open source)	JetBrains	Microsoft
Cost	Free	Community edition free	Free
Works in	Web browser	Desktop app	Desktop app
How code is written	In small blocks called cells; output appears below each cell	In full `.py` files	In full `.py` files
Best for	Learning, data analysis, graphs	Large projects	All-round use; light and fast
How to get it	Comes with Anaconda, or `pip install notebook`	Download from jetbrains.com	Download from code.visualstudio.com and add the Python extension

D. Steps to run your first program in Jupyter Notebook

1. Open Anaconda Navigator and click Launch under Jupyter Notebook (or type `jupyter notebook` in Command Prompt).
2. Jupyter opens in your web browser. Click New, then Python 3 to create a new notebook.
3. Type `print("Hello")` in the first cell.
4. Press Shift + Enter to run the cell. The output appears just below it.